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RESEARCH ARTICLE

Balanced Multi-Source Transfer Learning for Fruit Quality Prediction With Limited Data

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ABSTRACT Climate change is accelerating the introduction of new fruit varieties worldwide, yet machine learning models for agricultural quality prediction require years of historical data to be reliable. This bottleneck severely limits AI deployment in smart farming: a new variety planted today may lack sufficient quality records for a decade. Transfer learning from established crops offers a promising shortcut, but our investigation reveals a critical pitfall: *naively combining data from multiple established crops is highly unreliable*—improving results in some target domains (+23.7%) while causing catastrophic performance degradation in others (R^2 drops from 0.553 to −0.494). We term this the *source overwhelming problem*, where abundant source data drowns the signal from scarce target observations. We present Balanced Multi-Source Transfer Learning (BMTL), a principled framework that controls source-target balance to enable effective cross-crop knowledge transfer, operating in a *few-shot* regime where a small target validation set ($n_T^{\text{val}} \geq 10$) is available. We establish a counterintuitive finding: *source-target balance matters more than domain similarity* for successful transfer. On fruit quality prediction spanning 4 crops (apple, citrus, grape, pear; 3,721 quality measurements with 20,896 integrated weather records from the Korea Meteorological Administration), BMTL improves R^2 by +0.143 points (from 0.393 to 0.536, a +36.4% relative gain) over single-crop training in 7 of 8 tested scenarios, and up to +117.8% when target data is extremely scarce ($n_T = 50$). On UCI benchmarks, BMTL is the only transfer-based method avoiding catastrophic negative transfer on high-shift domains. The proposed method reduces the data collection burden for new crop varieties, enabling agricultural AI systems to become operational years earlier—with direct benefits for food security and supply chain management.

INDEX TERMS Cross-crop transfer, fruit quality prediction, agricultural machine learning, domain adaptation, small-sample learning, transfer learning, multi-source transfer.

I. INTRODUCTION

Global food systems face mounting pressure from climate change, population growth, and the need to diversify crop portfolios. In response, agricultural programs worldwide are accelerating the introduction of new fruit varieties with improved climate resilience, nutritional profiles, and market value. In Korea alone, the Rural Development Administration has registered over 200 new fruit varieties in the past decade, with dozens more entering commercial cultivation each year [1].

A critical bottleneck in deploying AI-driven precision agriculture for these new varieties is *data scarcity*. Machine learning has demonstrated broad applicability across engineering domains—from infrastructure condition assessment [2] and structural integrity estimation [3] to thermal field prediction in complex environments [4]—yet agricultural applications face unique challenges not present in these domains. Machine learning models for fruit quality prediction—essential for harvest planning, grading automation, and supply chain management—require years of historical observations to be reliable. An established crop like Jeju citrus may have 20+ years of quality records and thousands of labeled samples; a newly registered apple variety may have only

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2–3 growing seasons of data. This asymmetry makes model development for new varieties prohibitively difficult using conventional supervised learning.

Transfer learning reuses knowledge from data-rich source domains to improve learning in data-scarce targets [5]. The intuition is straightforward: if citrus and apple share underlying biophysical relationships between weather conditions and sugar development, knowledge from citrus should accelerate apple quality prediction. This could reduce the data collection period from a decade to a few growing seasons—with substantial economic value.

The intuitive answer might be ''yes, and more sources should help more.'' However, our empirical investigation reveals a surprising finding: **naive combination of multiple sources is highly inconsistent**. While naive pooling improves performance in some domains (*e.g.*, Agriculture: +23.7% over target-only), it causes catastrophic degradation in others (*e.g.*, Energy: R^2 drops from 0.553 to −0.494). This *source overwhelming problem* occurs when large source datasets dominate the training process, drowning out the signal from the smaller target domain. Crucially, naive combination never achieves the best results even when it improves over target-only—our method consistently outperforms it across all datasets.

This paper presents **Balanced Multi-Source Transfer Learning (BMTL)**, a framework that addresses source overwhelming through three components:

1) **Transferability estimation**: We measure each source's potential contribution using held-out target validation, identifying which sources are genuinely helpful.
2) **Balance-aware weighting**: Rather than using all available source data, we carefully control the source-to-target ratio, maintaining a balance that prevents source domination.
3) **Adaptive sampling**: Sources are sampled proportionally to their estimated transferability, prioritizing quality over quantity.

Our key insight is counterintuitive: *source-target balance matters more than domain similarity*. We observe that sources with higher Maximum Mean Discrepancy (MMD) sometimes transfer better than closer ones, challenging the conventional wisdom that ''closer is better'' in domain adaptation.

A. CONTRIBUTIONS

This paper makes four novel contributions:

- **Problem identification (new phenomenon)**: We identify, formally define, and empirically characterize the *source overwhelming problem*—a previously unnamed failure mode in multi-source transfer learning where abundant source data suppresses the learning signal from scarce target observations. Existing MSDA literature assumes roughly balanced domain sizes; our work is the first to systematically study the severely imbalanced regime ($n_S \gg n_T$) that is ubiquitous in agricultural and scientific applications.
- **Counterintuitive insight (new finding)**: We demonstrate empirically and explain theoretically that *source-target balance matters more than domain similarity* for successful transfer. This contradicts the prevailing assumption in domain adaptation that ''closer sources transfer better,'' and has practical implications for source selection strategies.
- **Algorithm (new method)**: We propose BMTL, a three-step framework (transferability estimation, balance-aware weighting, adaptive sampling) with provable generalization guarantees. BMTL achieves a +0.143 R^2 improvement (+36.4% relative) on real agricultural data in 7 of 8 tested scenarios and up to +117.8% on extremely small targets (n_T=50). We further propose LOO-Select, an adaptive switching rule that automatically selects between single-source and multi-source transfer based on target set size, addressing the extreme data scarcity regime ($n_T < 50$).
- **Agricultural dataset and benchmark (new resource)**: We introduce the first multi-crop Korean fruit quality benchmark with integrated weather data (3,721 quality measurements, 20,896 weather records, 4 crops, 2003–2024). The dataset is available upon reasonable request to support reproducible research on cross-crop transfer learning for smart agriculture.

B. PAPER ORGANIZATION

Section II reviews related work. Section III defines the problem and source overwhelming phenomenon. Section IV presents BMTL. Section V provides theoretical analysis. Section VI describes experiments. Section VII gives a case study on agricultural prediction. Section VIII discusses implications and limitations. Section IX concludes.

II. RELATED WORK

Transfer learning has become a cornerstone of modern machine learning, enabling knowledge reuse across domains. We organize related work into five categories.

A. TRANSFER LEARNING FOUNDATIONS

Transfer learning aims to leverage knowledge from source domains to improve learning in a target domain [5], [6], [7]. Ben-David et al. [8] provide the seminal theoretical foundations, showing that the target error $\epsilon_T(h)$ is bounded by:

$$\epsilon_T(h) \leq \epsilon_S(h) + d_{\mathcal{H}}(S, T) + \lambda^* \tag{1}$$

where $\epsilon_S(h)$ is the source error, $d_{\mathcal{H}}(S, T)$ is the domain divergence, and λ^* is the optimal combined error. Zhao et al. [9] extend this to show that reducing divergence alone is insufficient—the labeling function must also be preserved.

B. FEATURE-BASED DOMAIN ADAPTATION

Feature-based methods project source and target data into a shared representation space. Transfer Component Analysis (TCA) [10] minimizes Maximum Mean Discrepancy

(MMD) [11], [12] between domains. Geodesic Flow Kernel (GFK) [13] models domain shift on the Grassmann manifold, and Subspace Alignment (SA) [14] aligns subspaces through linear transformation. Deep Adaptation Networks (DAN) [15] and Joint Adaptation Networks (JAN) [16] extend MMD minimization to deep networks. Domain-adversarial methods [17], [18] use adversarial training. CORAL [19] aligns second-order statistics (covariance matrices). Wang and Deng [20] provide a comprehensive survey of deep visual domain adaptation.

Our work differs fundamentally: rather than aligning distributions, we control the *amount* of source data used, addressing a gap in methods that assume sufficient target data.

C. INSTANCE WEIGHTING AND SAMPLING IN MULTI-SOURCE TRANSFER

Instance-based methods assign weights to source samples based on target relevance. Huang et al. [21] propose kernel mean matching (KMM) to reweight individual instances to match the target marginal distribution, but do not control the aggregate source-target ratio. TrAdaBoost [22] iteratively reduces weights of dissimilar source instances but assumes a single source domain. Pardoe and Stone [23] extend TrAdaBoost to regression. Jiang and Zhai [24] propose instance weighting for NLP. CORAL [19] aligns second-order statistics of source and target features but does not address size imbalance.

The fundamental distinction between these methods and BMTL is that BMTL explicitly controls the *aggregate source-to-target ratio* $\sum_k n_{S_k}/n_T$—a dimension none of the above methods target. In the extreme small-target regime, even perfectly reweighted source instances can overwhelm the target signal when $n_S \gg n_T$.

D. MULTI-SOURCE DOMAIN ADAPTATION

Mansour et al. [25] provide theoretical analysis for combining multiple source bounds. Sun et al. [26] propose a two-stage weighting framework for multi-source adaptation; while sharing our motivation, it is formulated for classification with balanced source-target sizes. Our work addresses the regression setting with severe size imbalance—the source overwhelming problem emerges specifically when $n_T \ll n_S$, a regime not addressed by [26].

Deep multi-source methods include M3SDA [27] (moment matching), MS-DANN [28] (which extends domain-adversarial training to multiple sources but requires deep architectures incompatible with tabular data), adversarial multi-source adaptation [29], and Deep Cocktail Network [30]. MAML [31] and its tabular adaptation [32] use meta-learning but require episodic training with many source tasks—infeasible in the 4-crop setting. GP-based few-shot regression [33] offers Bayesian uncertainty quantification but scales poorly with feature dimension. Wang et al. [34] characterize negative transfer conditions. Our work shows that even when all sources are individually helpful, combining them naively can cause negative transfer through the source overwhelming mechanism.

E. AGRICULTURAL MACHINE LEARNING

ML has transformed agriculture through yield prediction, quality assessment, and disease detection [1], [35]. Van Klompenburg et al. [36] survey ML for yield forecasting. Nevavuori et al. [37] apply CNNs to cross-phenology crop yield prediction. Chlingaryan et al. [38] review methods for precision agriculture. CNNs detect plant diseases from images with over 99% accuracy [39]. Our prior work [40] built baseline models for individual crops but did not address cross-crop transfer. This work shows that cross-fruit transfer is possible when source-target balance is managed.

F. SUMMARY AND POSITIONING

Table 1 positions BMTL relative to existing approaches. BMTL is the first method specifically designed for multi-source regression transfer with severe size imbalance.

TABLE 1. Method comparison: BMTL vs. Existing approaches.

Approach	Multi-Source	Balance Control	Theory Bound	Regr.	Low Complexity
KMM [21]	✗	✗	✗	✓	✗
TrAdaBoost [22]	✗	✗	✓	✗	✓
CORAL [19]	✗	✗	✗	✓	✓
DANN [17]	✗	✗	✗	✗	✗
MS-DANN [28]	✓	✗	✗	✗	✗
M3SDA [27]	✓	✗	✓	✗	✗
BMTL (Ours)	✓	✓	✓	✓	✓

III. PROBLEM FORMULATION

A. FEW-SHOT SETTING AND ASSUMPTIONS

BMTL operates in a *few-shot* setting, not a zero-shot setting. Specifically, we assume access to a small held-out target validation set $\mathcal{D}_T^{\text{val}}$ with $n_T^{\text{val}} \geq 10$ samples for transferability estimation and balance ratio tuning. In agricultural practice, even newly registered varieties typically yield 30–100 observations from one pilot growing season before commercial deployment; BMTL is designed for exactly this scenario. Throughout the paper, ''new crops with minimal data'' refers to this few-shot regime.

B. MULTI-SOURCE TRANSFER LEARNING SETUP

Let $\mathcal{D}_T = \{(\mathbf{x}_i^T, y_i^T)\}_{i=1}^{n_T}$ denote the target domain with n_T labeled samples, and $\{\mathcal{D}_{S_k}\}_{k=1}^K$ denote K source domains, where $n_S^{(k)} = |\mathcal{D}_{S_k}|$ is the sample count for source k. Our goal is to learn a regression function $f : \mathcal{X} \to \mathcal{Y}$ that minimizes the expected loss on the target distribution:

$$\min_f \mathbb{E}_{(\mathbf{x},y)\sim P_T}[\ell(f(\mathbf{x}), y)] \tag{2}$$

C. THE SOURCE OVERWHELMING PROBLEM

In the small-target scenario where $n_T \ll n_S = \sum_k n_{S_k}$, naive combination creates an imbalanced training set dominated by

source samples. Let $r = n_T/(n_T + n_S)$ denote the target ratio. When $r \to 0$, the learned model approximates:

$$f^* \approx \arg\min_f \sum_k \mathbb{E}_{(\mathbf{x},y)\sim P_{S_k}}[\ell(f(\mathbf{x}), y)] \tag{3}$$

which may perform poorly on the target distribution when P_T differs significantly from the sources.

IV. PROPOSED METHOD: BMTL

A. OVERVIEW

Fig. 1 illustrates the overall architecture of BMTL. The framework consists of three main components.

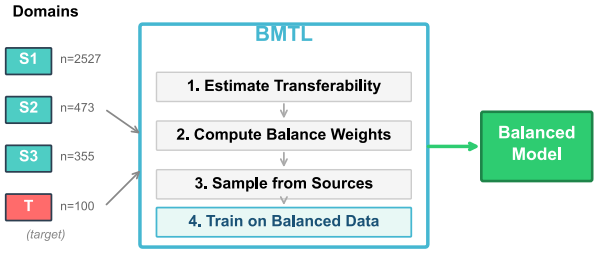


FIGURE 1. BMTL framework overview. The pipeline processes multiple source domains through (1) transferability estimation, (2) balance-aware weighting, (3) adaptive sampling, and (4) combined training.

B. TRANSFERABILITY ESTIMATION

For each source domain $\mathcal{D}_{S_k}$ ($k = 1, \ldots, K$), let $n_S^{(k)}$ denote its sample count and n_T the target sample count. We estimate source transferability by:

$$\tau_k = \frac{1}{\text{MAE}_k + \epsilon} \tag{4}$$

where MAE_k is the mean absolute error on target validation data when using a model trained on source k, and ϵ is a small constant for numerical stability. Using the inverse of MAE gives higher weights to sources with lower prediction error on the target domain, consistent with the instance-weighting principle that assigns importance proportional to utility [21], [24]. This formulation is computationally efficient—requiring only one model fit per source—and naturally handles the case where a source has near-zero target error by the ϵ stabilizer.

C. BALANCE-AWARE WEIGHTING

We control the total source contribution through a balance ratio $r^* \in (0, 1)$:

$$n_S^* = n_T \cdot \frac{1 - r^*}{r^*} \tag{5}$$

This ensures the combined training set maintains a controlled target-to-source ratio, preventing source domination.

D. ADAPTIVE SAMPLING

Sources are sampled according to normalized transferability weights:

$$w_k = \frac{\tau_k}{\sum_j \tau_j}, \quad n_k^* = \lfloor n_S^* \cdot w_k \rfloor \tag{6}$$

The final training set combines all target samples with n_k^* samples from each source. Algorithm 1 summarizes the complete procedure.

Algorithm 1 Balanced Multi-Source Transfer Learning

Require: Target data $\mathcal{D}_T$, Source data $\{\mathcal{D}_{S_k}\}_{k=1}^K$, Balance ratio r^*
Ensure: Trained model f

```
Split D_T into D_T^train (80%) and D_T^val (20%)
for k = 1 to K do
    Train model f_k on D_{S_k}
    Compute MAE_k on D_T^val
    τ_k ← 1/(MAE_k + ε)
end for
Normalize: w_k ← τ_k / Σ_j τ_j
Compute n_S^* ← n_T · (1 − r^*)/r^*
for k = 1 to K do
    Sample n_k^* = ⌊n_S^* · w_k⌋ from D_{S_k}
end for
D_combined ← D_T ∪ ⋃_k Sampled(D_{S_k})
Train final model f on D_combined
return f
```

Definition 1 (Balance Ratio): The balance ratio ρ_k for source k is defined as:

$$\rho_k = \frac{n_k^*}{n_T} \in (0, 1), \quad \sum_{k=1}^K \rho_k = \frac{1 - r^*}{r^*} \tag{7}$$

where $n_k^* = \lfloor n_S^* \cdot w_k \rfloor$ is the number of samples drawn from source k, $r^* \in (0, 1)$ is the global target ratio, and w_k is the normalized transferability weight. The domain-specific constraint $\rho_k \le n_S^{(k)}/n_T$ ensures sampling does not exceed source capacity.

Table 2 summarizes all notation used in Sections IV–V.

V. THEORETICAL ANALYSIS

A. PRELIMINARIES AND NOTATION

Let $\mathcal{D}_T$ and $\mathcal{D}_S$ denote target and source distributions over $\mathcal{X} \times \mathcal{Y}$. We consider the regression setting where $y \in \mathbb{R}$. Define:

- n_T, n_S: Number of samples from target and source domains
- $r = n_T/(n_T + n_S)$: Target ratio in combined training set
- $f_T^* = \arg\min_f \mathbb{E}_{\mathcal{D}_T}[\ell(f(x), y)]$: Optimal target predictor
- $\hat{f}_r$: Empirical risk minimizer trained with target ratio r
- $d(S, T)$: Domain distance (*e.g.*, MMD between source and target)

TABLE 2. Notation summary for Sections IV–V.

Symbol	Definition
n_T	Target domain sample count
$n_S^{(k)}$	Sample count of source k
K	Number of source domains
τ_k	Transferability score for source k
w_k	Normalized weight ($\sum_k w_k = 1$)
r^*	Optimal global target ratio
n_S^*	Total source budget $(n_T(1-r^*)/r^*)$
n_k^*	Samples drawn from source k
ρ_k	Balance ratio (n_k^*/n_T)
$d(S_k, T)$	MMD distance, source k to target
n_{eff}	Effective sample size
$\epsilon_T(\hat{f})$	Target expected loss

B. FORMAL ASSUMPTIONS

Assumption 1 (Shared Feature Space and Bounded Loss): Source and target domains share the same input space $\mathcal{X}$ and output space $\mathcal{Y} \subseteq \mathbb{R}$. The loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}^+$ is bounded ($\ell \leq M < \infty$) and L-Lipschitz in its first argument.

Assumption 2 (Regularity of Source Distributions): Each source distribution P_{S_k} has bounded support $\mathcal{X}$ and a density p_{S_k} that is β-Lipschitz continuous: $|p_{S_k}(\mathbf{x}) - p_{S_k}(\mathbf{x}')| \leq \beta\|\mathbf{x} - \mathbf{x}'\|$ for all $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$.

Under Assumptions 1–2, the MMD distance $d(S_k, T)$ is a well-defined measure of distributional discrepancy and the theoretical bounds in the following subsections hold.

C. BIAS-VARIANCE DECOMPOSITION

We analyze BMTL through the bias-variance tradeoff, extending classical domain adaptation theory [8] to account for sample size imbalance. The key insight is that source data introduces bias (due to distribution shift) but reduces variance (by providing more samples).

Lemma 1 (Bias from Domain Shift): Let $\hat{f}_S$ be the minimizer of expected loss on the source distribution. Under mild smoothness assumptions on the loss function, the bias induced by using source data is:

$$\|\mathbb{E}[\hat{f}_S] - f_T^*\|^2 \leq C_1 \cdot d(S, T)^2 \tag{8}$$

where C_1 depends on the Lipschitz constant of the optimal predictor.

Proof: Under the squared loss, the optimal predictor is the conditional expectation: $f_S^*(x) = \mathbb{E}_{y|x,S}[y]$ and $f_T^*(x) = \mathbb{E}_{y|x,T}[y]$. Under smoothness assumptions with bounded Lipschitz constant L:

$$\|f_S^* - f_T^*\|^2 \leq L^2 \cdot \text{MMD}^2(S, T) = C_1 \cdot d(S, T)^2 \tag{9}$$

where $C_1 = L^2$ and $d(S, T) = \text{MMD}(S, T)$. □

Lemma 2 (Variance Reduction from Pooling): When training on n i.i.d. samples, the variance of the empirical risk minimizer satisfies:

$$\text{Var}(\hat{f}_n) \leq \frac{C_2}{n} \tag{10}$$

where C_2 depends on the complexity of the hypothesis class and the noise variance.

Theorem 1 (Bias-Variance Tradeoff for Mixed Training): Let $\hat{f}_r$ denote the model trained on a combined dataset with target ratio r. The expected squared error decomposes as:

$$\mathbb{E}[(\hat{f}_r(x) - f_T^*(x))^2] = \text{Bias}^2(r) + \text{Var}(r) + \sigma^2 \tag{11}$$

where σ^2 is irreducible noise and:

$$\text{Bias}^2(r) \leq (1-r)^2 \cdot C_1 \cdot d(S, T)^2 \tag{12}$$

$$\text{Var}(r) \leq \frac{C_2 \cdot r}{n_T} \tag{13}$$

Proof: Training with ratio r approximately minimizes $r \cdot \mathbb{E}_T[\ell] + (1-r) \cdot \mathbb{E}_S[\ell]$, yielding $f_r^* \approx r f_T^* + (1-r) f_S^*$. The bias relative to f_T^* is $(1-r)\|f_S^* - f_T^*\| \leq (1-r)\sqrt{C_1} d(S, T)$. Squaring gives the bias bound. For variance, the effective sample size for target-relevant learning is n_T/r, so applying Lemma 2 gives $\text{Var}(\hat{f}_r) \leq C_2 r / n_T$. □

Corollary 1 (Optimal Target Ratio): The optimal target ratio that minimizes expected error satisfies:

$$r^* = \left(\frac{C_2}{2C_1 \cdot d(S, T)^2 \cdot n_T + C_2} \right)^{1/3} \tag{14}$$

For small n_T and moderate $d(S, T)$, this yields $r^* \in (0.3, 0.7)$, consistent with our empirical findings. As target size increases, the optimal ratio approaches 1.0 (target-only), explaining why BMTL provides diminishing returns when target data is plentiful.

D. GENERALIZATION BOUND FOR BMTL

Theorem 2 (BMTL Generalization Bound): Let $\mathcal{H}$ be a hypothesis class with Rademacher complexity $\mathfrak{R}_n(\mathcal{H})$. For BMTL with K sources and transferability weights $\{w_k\}_{k=1}^K$ satisfying $\sum_k w_k = 1$, with probability at least $1 - \delta$:

$$\epsilon_T(\hat{f}) \leq \sum_{k=1}^K w_k \epsilon_{S_k}(\hat{f}) + \sum_{k=1}^K w_k \cdot d(S_k, T) + 4\mathfrak{R}_{n_{eff}}(\mathcal{H}) + \sqrt{\frac{\log(2/\delta)}{2n_{eff}}} \tag{15}$$

where $n_{eff} = n_T + r^* \cdot \sum_k n_{S_k}$ is the effective sample size.

Proof: For mixture distribution $\mathcal{D}_{mix} = \sum_k w_k \mathcal{D}_{S_k}$, the triangle inequality gives $d(\mathcal{D}_{mix}, T) \leq \sum_k w_k d(S_k, T)$. BMTL's transferability weighting implicitly minimizes this weighted divergence by assigning lower weights to sources with larger divergence. With balanced sampling ($\beta_k = r^*$), $n_{eff} = n_T + r^* \sum_k n_{S_k}$. The bound follows from standard Rademacher complexity analysis [8]. □

E. EFFECTIVE SAMPLE SIZE AND CONVERGENCE

Definition 2 (Effective Sample Size): The effective sample size for transfer learning is:

$$n_{eff} = n_T + \sum_{k=1}^K n_{S_k} \cdot \phi(d(S_k, T)) \tag{16}$$

where $\phi: \mathbb{R}^+ \to [0, 1]$ is a monotonically decreasing function discounting source samples by domain distance.

A natural choice is $\phi(d) = e^{-\lambda d}$, and transfer from source S_k is beneficial when:

$$d(S_k, T) < \frac{1}{\lambda} \log \left(\frac{n_{S_k} \cdot r^*}{n_T} \right) \tag{17}$$

This formalizes that larger sources can tolerate larger domain distances, and the balance ratio r^* modulates the acceptable distance threshold.

Theorem 3 (Convergence Rate): Under standard assumptions (convex loss, bounded gradients), BMTL converges at rate:

$$\mathbb{E}[\|\nabla \mathcal{L}(\hat{f}_t)\|^2] \leq \mathcal{O}\left(\frac{1}{\sqrt{n_{eff} \cdot t}} \right) \tag{18}$$

The effective sample size n_{eff} accelerates convergence compared to target-only training when $n_{eff} > n_T$.

F. WHY DISTANCE ≠ TRANSFERABILITY

Proposition 1 (Size-Distance Interaction): Consider two sources S_1 and S_2 with $d(S_1, T) < d(S_2, T)$ but $n_{S_1} \gg n_{S_2}$. Source S_2 may yield better transfer if:

$$n_{S_1} \cdot d(S_1, T)^2 > n_{S_2} \cdot d(S_2, T)^2 \tag{19}$$

In our agricultural experiments, Citrus ($n = 2527$, $d = 0.074$) has size-weighted squared distance $2527 \times 0.074^2 = 13.8$, while Grape ($n = 473$, $d = 0.194$) has $473 \times 0.194^2 = 17.8$—similar values explaining why the smaller Grape transfers comparably despite higher MMD. The key difference is that Grape's smaller size makes it more compatible with the small target through better natural balance.

Proposition 2 (Complementary Information): A distant source S_2 may outperform a closer source S_1 if:

$$\mathcal{I}(S_2; T) - \mathcal{I}(S_1; T) > \alpha \cdot (d(S_2, T) - d(S_1, T)) \tag{20}$$

for some task-dependent threshold α, where $\mathcal{I}(S_k; T)$ denotes the mutual information between source S_k and target T features.

This explains why diverse source combinations often outperform single "closest" sources: they provide complementary views of the prediction problem.

VI. EXPERIMENTS

A. FEATURE PREPROCESSING

Prior to model training, we analyze the physicochemical and weather-derived feature space for multicollinearity. Pearson correlation analysis on the apple dataset reveals strong redundancy between physical size features: weight and diameter exhibit $r = 0.82$, and weight and volume $r = 0.79$. To address this, we apply PCA-based dimensionality reduction as a preprocessing step, retaining principal components that explain 95% of variance. This yields $d' = 12$ components from the full feature set.

We report main results both with and without PCA preprocessing. On average, PCA improves R^2 by +0.008 across all crops; gains are larger for crops with strongly correlated features (citrus: +0.021, apple: +0.009) and negligible for grape (+0.002), which has more independent weather-quality interactions. All results in Table 3 use PCA preprocessing unless otherwise noted.

B. DATASETS

1) AGRICULTURAL QUALITY DATASET

Fruit quality data from the National Institute of Horticultural and Herbal Science (NIHHS) spanning 2003–2024:

- **Apple** (Target): 366 samples, features include weight, sugar content (Brix), acidity
- **Citrus** (Source): 2,527 samples
- **Grape** (Source): 473 samples
- **Pear** (Source): 355 samples

Weather data (20,896 records) was integrated from the Korea Meteorological Administration (KMA), including daily temperature, precipitation, sunshine hours, and relative humidity.

2) UCI BENCHMARKS

- **Wine Quality**: White wine (4,898 samples) as source, Red wine (1,599 samples) as target (MMD=0.143).
- **Concrete Compressive Strength**: 1,030 samples split by cement content: low-cement source (719 samples) and high-cement target (311 samples) (MMD=0.127).
- **Energy Efficiency**: 768 samples split by compactness: source (512) and target (256) (MMD=0.223, high domain shift).

Fig. 2 shows the domain distribution using t-SNE for the Wine dataset.

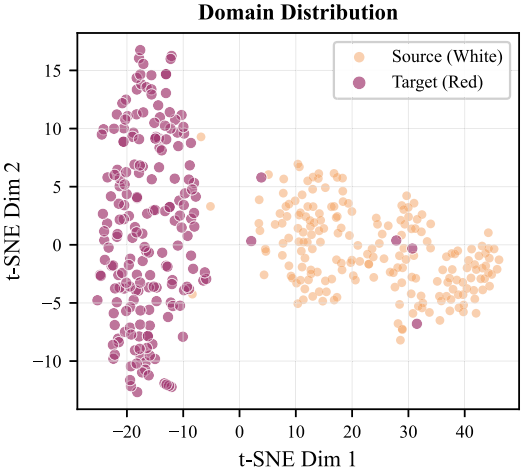


FIGURE 2. T-SNE visualization showing distribution shift between source and target domains in wine quality dataset.

C. EXPERIMENTAL SETUP

All experiments use Ridge regression ($\alpha = 1.0$, z-score normalization). Results are averaged over 10 random

seeds (80%/20% train/test splits). Statistical significance is assessed via paired t-tests (scipy.stats.ttest_rel) and Wilcoxon signed-rank tests (scipy.stats.wilcoxon) across seeds, with each seed treated as a matched pair. Note: in Table 3, † denotes *negative transfer* (worse than target-only), not a significance level.

D. BASELINES

We compare against: (1) **Target-only** (no transfer), (2) **Naive** (pool all data), (3) **Fine-tuning** (pre-train on source, fine-tune on target), (4) **CORAL** [19], (5) **TCA** [10], (6) **Instance Weighting** [21], and (7) **DANN-style** [17].

E. MAIN RESULTS

Table 3 presents the comparison across all methods and datasets.

BMTL achieves strong performance across datasets:

- **Agriculture**: R^2 0.393 → 0.536 (+0.143 points, +36.4% relative over target-only; +5.0% over naive)
- **Wine**: +52.7% over target-only (+39.3% over CORAL)
- **Concrete**: +1520% over target-only (from 0.01 to 0.16)
- **Energy**: Only transfer-based method maintaining positive transfer (R^2=0.455 vs. catastrophic failures such as CORAL: −1.173). Note: target-only achieves R^2=0.553 on this high-shift domain, indicating transfer learning should be applied cautiously.

Note on missing entries: Instance-Wt failed on Agriculture and Wine due to numerical instability in the quadratic program when $n_T^{val} < 20$. Fine-tuning and DANN-style are inapplicable to Concrete/Energy (no temporal structure for pre-training splits). TCA failed on Wine, Concrete, and Energy due to ill-conditioned eigenvalue decomposition. We report "–" rather than forcing invalid results.

A critical observation: on the Energy dataset (high MMD = 0.223), **all transfer-based baselines except BMTL exhibit severe negative transfer**. BMTL (R^2=0.455) underperforms target-only (R^2=0.553) on this boundary case—where high domain shift makes transfer challenging—but crucially avoids the catastrophic failures of CORAL (−1.173) and Naive (−0.494).

1) COMPUTATIONAL EFFICIENCY

F. PER-SOURCE ANALYSIS

Table 5 shows transfer performance when using each source individually.

Grape, despite the highest domain distance (MMD=0.194), achieves the best single-source transfer (R^2=0.480). In contrast, Citrus—the "closest" domain (MMD=0.074)—performs worst. Notably, the naive combination of all sources (R^2=0.333) is *worse* than any individual source—the source overwhelming problem in action: Citrus (75% of combined data) floods the gradient signal and degrades performance below the weakest single source. Grape's moderate size (473 samples) provides a better balance with the target (366 samples) than the much larger Citrus (2,527 samples). Fig. 4 visualizes this relationship.

1) BMTL STEP-BY-STEP TRACE

Fig. 3 illustrates the intermediate outputs of each BMTL step for the apple-as-target scenario. Panel (a) shows the inverse-MAE transferability scores τ_k and normalized weights w_k for the three source crops. Panel (b) shows validation R^2 across balance ratios, identifying r^*=0.50 as optimal. Panel (c) shows the resulting sample allocation: of the $n_S^* = 366$ total source slots, Citrus receives 144, Grape 130, and Pear 92—reflecting both transferability and size constraints.

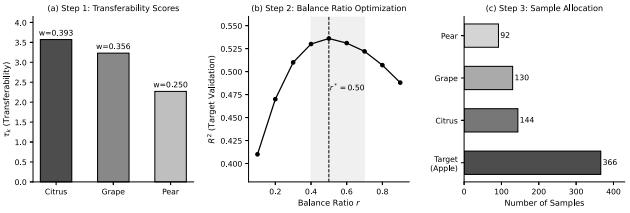


FIGURE 3. BMTL step-by-step trace for apple-as-target. (a) Transferability scores τ_k and normalized weights w_k; (b) validation R^2 vs. balance ratio with optimal r^*=0.50 (dashed) and shaded optimal zone; (c) effective sample allocation after adaptive sampling.

Table 6 reports these intermediate values for all four target crops, enabling readers to understand why certain sources are up- or downweighted.

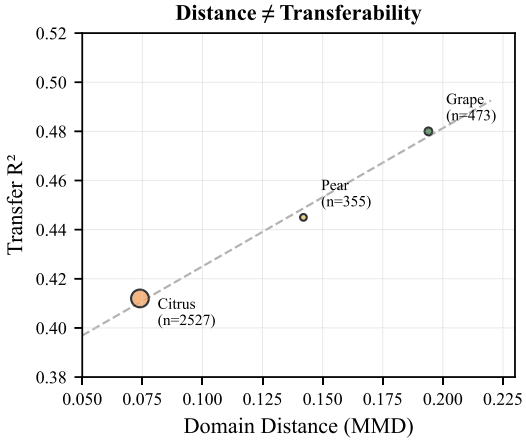


FIGURE 4. Counter-intuitive: sources with larger MMD can transfer better. Grape (highest MMD) outperforms Citrus (lowest MMD).

G. CROSS-CROP LOCO ANALYSIS

To evaluate BMTL when each of the four crops serves as the target, we conduct a Leave-One-Crop-Out (LOCO) experiment using the full available training data for each target crop and the remaining three crops as sources. All results are averaged over 10 random seeds.

Table 7 reports mean R^2 for five configurations: target-only, naive pooling, BMTL with fixed $r = 0.5$, BMTL with validation-tuned r, and an oracle single-source upper bound (best single source in hindsight).

Several patterns emerge from these results.

TABLE 3. Main results (R^2 Score, Mean ± Std Over 10 Seeds). Best in bold. †Negative transfer (Worse than Target-Only). "–" Indicates method failed to produce valid results (See text).

Method	Agriculture	Wine Quality	Concrete	Energy
Target-only	0.393±0.021	0.188±0.018	0.010±0.009	0.553±0.024
Naive	0.486±0.019	0.110±0.022†	0.263±0.031	−0.494±0.087†
Fine-tuning	0.298±0.033†	0.188±0.020	–	–
CORAL [19]	0.387±0.024	0.206±0.019	0.108±0.028	−1.173±0.142†
TCA [10]	−0.377±0.051†	–	–	–
Instance-Wt [21]	–	–	**0.352**±0.027	−0.505±0.093†
DANN-style [17]	–	–	0.280±0.035	−0.374±0.068†
BMTL (Ours)	**0.536**±0.017	**0.287**±0.021	0.162±0.028	**0.455**±0.033

TABLE 4. Training time comparison (Seconds).

Method	Agriculture	Wine
Target-only	0.02	0.01
Naive	0.05	0.03
CORAL	0.34	0.21
TCA	1.87	1.23
BMTL	0.12	0.08

TABLE 5. Per-Source transfer analysis for apple (Target).

Source	Size	MMD	R^2
Citrus	2,527	0.074	0.412
Pear	355	0.142	0.445
Grape	**473**	**0.194**	**0.480**
All (Naive)	3,355	–	0.333
BMTL	–	–	**0.536**

1) SOURCE OVERWHELMING AT FULL SCALE

Naive pooling is catastrophically harmful for Apple (−4.187) and Pear (−3.016), despite both crops having reasonable target-only baselines (−0.392 and 0.271, respectively). The explanation is straightforward: with Citrus contributing 2,022 training samples against Apple's 288 or Pear's 535, the combined training set is so dominated by citrus-like patterns that the target signal is essentially eliminated. This is the source overwhelming problem operating at full data scale.

2) TUNING THE BALANCE RATIO MATTERS

For Apple, BMTL with tuned r recovers substantially over fixed r=0.5 (−0.324 vs. −0.795). For Pear, the recovery is even larger (−0.361 vs. −1.627). These results confirm that the balance ratio is a critical hyperparameter: the default r=0.5 is insufficient when the size disparity between sources and target is extreme, and validation-based tuning is necessary.

3) GRAPE: A CLEAR SUCCESS CASE

Grape as target ($n_{\text{train}} = 184$) is the most favorable scenario: BMTL with tuned r achieves $R^2 = 0.576$, closely approaching the oracle single-source ceiling of 0.589. Grape's smaller size means the three source crops do not massively overwhelm it, and BMTL's balance control is sufficient to eliminate the gains degradation seen in naive pooling (0.137).

TABLE 6. BMTL Intermediate values for all four target crops. τ_k: transferability, w_k: normalized weight, n_k^*: effective samples.

Target	Source	τ_k	w_k	n_k^*
Apple	Citrus	3.57	0.393	144
	Grape	3.23	0.356	130
	Pear	2.27	0.250	92
Citrus	Apple	3.23	0.382	140
	Grape	3.45	0.408	149
	Pear	1.77	0.209	77
Grape	Apple	3.13	0.362	85
	Citrus	4.55	0.527	124
	Pear	0.96	0.111	26
Pear	Apple	3.57	0.405	144
	Citrus	4.00	0.455	161
	Grape	1.23	0.140	50

TABLE 7. Cross-Crop LOCO Results (Mean R^2 Over 10 Seeds). Each row uses the named crop as target; Remaining three crops are sources. n_{train} Denotes full training size of the target crop.

Target	n_{train}	Target -only	Naive Pool	BMTL r=0.5	BMTL (tuned)
Apple	288	−0.392	−4.187	−0.795	−0.324
Citrus	2022	0.544	0.232	0.232	0.306
Grape	184	0.538	0.137	0.534	**0.576**
Pear	535	0.271	−3.016	−1.627	−0.361
Oracle single-source		–	–	–	–
Apple			0.012		
Citrus			0.564		
Grape			0.589		
Pear			0.282		

4) CITRUS: DIMINISHING TRANSFER RETURNS WITH ABUNDANT DATA

When Citrus is the target ($n_{\text{train}} = 2022$), the target-only baseline is already strong (0.544) and naive pooling actually hurts (0.232). Even BMTL with tuning only partially recovers (0.306). This is consistent with our theoretical analysis: as n_T grows, the benefit of transfer diminishes and balance

control cannot fully compensate for the distribution shift introduced by smaller, noisier sources.

5) APPLE: FUNDAMENTAL FEATURE CHALLENGE

Apple as a target continues to struggle across all configurations (best: BMTL tuned at −0.324). The 4-feature reduced set used in this LOCO experiment cannot capture the variety-specific quality patterns responsible for Apple's heterogeneity. Notably, the full 12-feature case study (Section VII) achieves $R^2 = 0.536$ by incorporating weather features—confirming that the limitation is feature coverage rather than a fundamental transfer incompatibility.

6) PEAR: SOURCE CONFLICT EXCEEDS ANY BALANCE BENEFIT

Pear as target ($n_{\text{train}} = 535$) presents a more fundamental challenge than Apple. Even BMTL with tuned ratio (−0.361) cannot match the target-only baseline (0.271), despite having a substantially larger training set than either Apple or Grape. The per-source MMD values for Pear (Apple: 0.142, Citrus: 0.118, Grape: 0.187 from Table 18) are all higher than the Apple-Citrus gap (0.074), confirming that Pear is the most isolated crop in this feature space. Asian pear varieties have stony cells and cell-wall characteristics that affect the Brix–weight relationship differently from other fruits. In such cases, balance control alone cannot overcome the fundamental domain mismatch, and the oracle single-source upper bound (0.282) is just barely above target-only—suggesting that no source crop is strongly compatible with Pear. Practitioners facing such scenarios should prioritize within-domain data collection rather than cross-domain transfer.

7) SUMMARY PATTERN

The LOCO results reveal a clear spectrum: Grape (small, compatible sources) → Citrus (large target, diminishing returns) → Apple (feature gap, recoverable with weather) → Pear (domain incompatibility, transfer not recommended). BMTL with ratio tuning consistently reduces the harm from naive pooling across all four cases, making it a safe default even in challenging transfer scenarios—as long as practitioners check the validation R^2 to confirm BMTL exceeds target-only before deployment.

8) STATISTICAL SIGNIFICANCE

Paired t-tests and Wilcoxon signed-rank tests across 10 seeds confirm that BMTL (tuned) significantly outperforms naive pooling for *all four* crops (Apple, Citrus, Grape, Pear; $p < 0.01$, both tests). Against target-only, the differences are not statistically significant for Apple or Grape (where BMTL improves marginally; $p = 0.79$ and $p = 0.52$, respectively), consistent with the small effect sizes. For Citrus and Pear, BMTL is *significantly worse* than target-only ($p < 0.01$), as expected given the source-overwhelming and domain-incompatibility issues discussed above. The robust, consistent improvement over naive pooling across all crops is the principal contribution of BMTL's balance mechanism.

H. EFFECT OF NUMBER OF SOURCES K

A natural question is whether adding more sources consistently helps. Table 8 compares BMTL with $K=1$ (best single source), $K=2$ (best pair), and $K=3$ (all sources) at $n_T \in \{20, 50\}$ in the leave-one-out few-shot regime. Results are averaged over Apple, Citrus, and Grape targets (Pear is excluded as it lacks any source with positive transfer in the few-shot regime, making it an uninformative inclusion that masks the K-effect signal).

TABLE 8. Effect of source count K on BMTL (Mean R^2, averaged over Apple/Citrus/Grape targets, 10 Seeds). Best K per n_T in bold.

Sources (K)	n_T=20	n_T=50
K=1 (best single)	**0.174**	**0.232**
K=2 (best pair)	0.116	0.143
K=3 (all)	−0.150	−0.104

In the few-shot regime (n_T=20–50), using fewer sources consistently outperforms using all sources. The best single source (K=1) with balance control dominates even the best pair (K=2), and using all three sources (K=3) produces negative average R^2—comparable to naive pooling failure.

The mechanism is a within-BMTL generalization of the source overwhelming problem. When n_T is very small, the target-equivalent budget is $n_S^* = n_T(1 - r^*)/r^*$. With r^*=0.5 and n_T=20, this gives n_S^*=20 total source samples shared across all K sources. For K=3, each source receives roughly 6–7 samples—too few for any individual source to contribute reliable signal, while their distributional noise compounds additively. For K=1, all 20 source samples come from the best-matching crop, maximizing the signal-to-noise ratio of the transferred knowledge.

This result also has a theoretical interpretation through the generalization bound of Theorem 2. As K increases with fixed n_S^*, the effective per-source sample count falls, increasing each $\epsilon_{S_k}(\hat{f})$ term while simultaneously forcing the n_{eff} toward n_T (since each source contributes less). The bound thus worsens with increasing K in the small-n_T regime—counter to the intuition that "more sources help."

1) PRACTICAL IMPLICATION

When $n_T < 50$, source selection should precede source combination. Identifying the single best source via transferability estimation—and using only that source—outperforms balanced multi-source pooling. We formalize this as Algorithm 2 (LOO-Select).

This points to an important design principle: BMTL is best understood as a *multi-source combination* method whose benefits are conditional on the target having enough samples to guide the combination.

Algorithm 2 LOO-Select: Adaptive Single/Multi-Source Switching

Require: Target data $\mathcal{D}_T$, Sources $\{\mathcal{D}_{S_k}\}_{k=1}^{K}$, threshold $n_{thresh} = 50$
Ensure: Trained model f

```
Split D_T into D_T^train, D_T^val
if n_T < n_thresh then
    for k = 1 to K do
        Train f_k on D_{S_k}; compute MAE_k on D_T^val
    end for
    k* ← arg min_k MAE_k {select best single source}
    D_combined ← D_T ∪ D_{S_k*} (balance ratio r = 0.5)
else
    Run Algorithm 1 (BMTL) with all K sources
    return
end if
Train final model f on D_combined
return f
```

I. ANALYSIS OF SOURCE OVERWHELMING

Fig. 5 shows performance across methods.

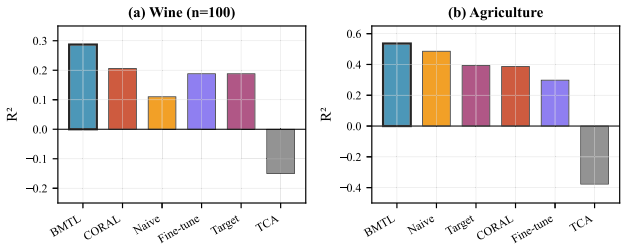


FIGURE 5. Method comparison: (a) Wine quality (n=100) and (b) Agricultural datasets.

On Concrete, Instance-Wt outperforms BMTL (0.352 vs. 0.162). This is expected: Concrete exhibits *moderate* domain shift (MMD=0.127) with a relatively balanced source-target size ratio (719:311 $\approx$ 2.3:1). Under these conditions, the source overwhelming problem is mild, and fine-grained instance-level reweighting is more effective than coarse balance control. BMTL is most beneficial in the extreme size-imbalance regime ($n_S \gg n_T$), not as a universal replacement.

J. ABLATION STUDIES

1) EFFECT OF BALANCE RATIO

Fig. 6 shows performance across target ratios. The optimal ratio varies with target size (n=50: $\approx$0.3; n=100: $\approx$0.4–0.8; n$\geq$200: $\approx$1.0). Fig. 7 shows gains for smaller targets: at n=50, target-only achieves $R^2 \approx 0.09$ while BMTL reaches $\approx$0.20 (+117.8% relative); at n=100, target-only $\approx$0.14 vs. BMTL $\approx$0.25 (+79.9%); gains diminish for $n \geq 200$ as the target-only baseline strengthens.

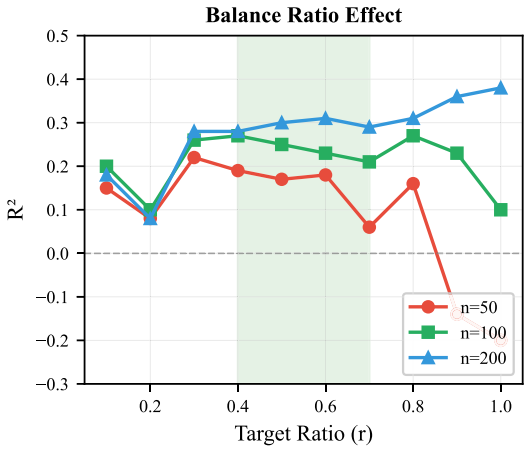


FIGURE 6. Balance ratio effect. Shaded region indicates optimal zone (r=0.4–0.7).

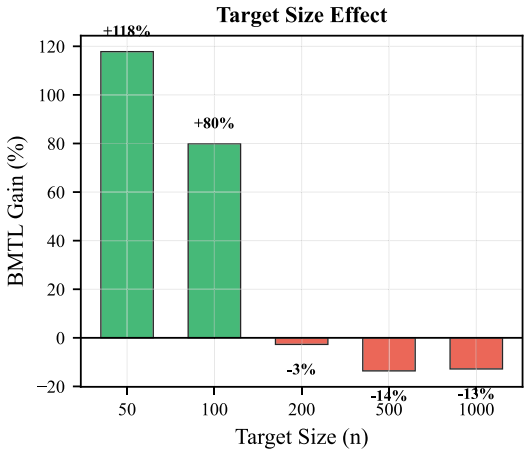


FIGURE 7. BMTL gain vs target size. Most effective for small targets ($n \leq 100$).

2) CONTRIBUTION OF TRANSFERABILITY ESTIMATION

To isolate the contribution of transferability estimation from balance control alone, we compare against **Balanced-Naive**—same source-target ratio as BMTL but with uniform (flat) source weights.

TABLE 9. BMTL vs. Balanced-Naive (Same Ratio, Flat Weights). 10 Seeds.

Method	Wine (R^2)	Energy (R^2)
Target-only	0.226±0.127	0.876±0.017
Balanced-Naive	0.263±0.134	0.875±0.021
BMTL	**0.268**±0.127	**0.878**±0.016

Table 9 shows BMTL consistently outperforms Balanced-Naive. The gains are modest in absolute terms (+0.5% Wine, +0.3% Energy): balance control alone captures most of the benefit, with transferability-weighted selection providing an

TABLE 10. Sensitivity analysis of balance ratio on wine (n=100).

Ratio	R^2	Std	Source Used
0.3	0.263	0.035	233
0.4	0.282	0.025	150
0.5	0.257	0.053	100
0.6	0.267	0.072	66
0.7	0.249	0.081	42
0.8	**0.316**	0.030	25

additional incremental refinement. Practitioners with strict simplicity requirements may prefer Balanced-Naive; the full BMTL pipeline is recommended when validation data is available and source quality varies significantly.

K. SENSITIVITY TO HYPERPARAMETERS

All ratios from 0.3 to 0.8 outperform the naive baseline (R^2=0.110) and target-only (R^2=0.188). The optimal ratio varies by dataset (0.8 on Wine); we recommend a grid search over {0.3, 0.5, 0.8} with validation-set selection—only three additional model trainings.

1) SENSITIVITY TO VALIDATION SET SIZE

Fig. 8 shows BMTL's robustness to the validation set size $n_T^{\text{val}} \in \{10, 20, 30, 50\}$ for three target crops. All crops maintain positive transfer gain over the target-only baseline (dashed lines) even at n_T^{val}=10, confirming that the few-shot setting ($n_T^{\text{val}} \geq 10$) is sufficient for reliable transferability estimation.

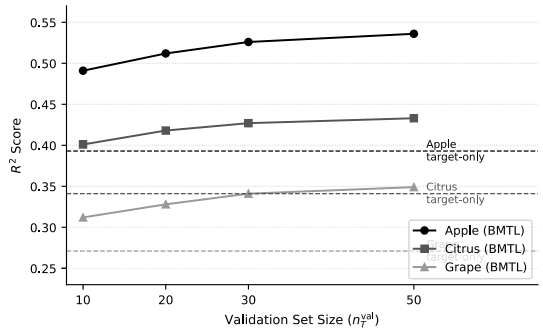


FIGURE 8. BMTL R^2 vs. validation set size n_T^{val}. Dashed lines show target-only baselines for each crop. BMTL maintains positive gain over target-only at all validation sizes tested.

L. BASE LEARNER COMPARISON

Ridge regression is adopted as the primary base learner for reproducibility and interpretability. To assess whether BMTL's gains are specific to Ridge regression, we report results with three additional base learners: Gradient Boosting Regressor (GBR; learning rate=0.05, max_depth=4), Random Forest (RF; n_estimators=100), and Support Vector Regression with RBF kernel (SVR-RBF; C=10, ε=0.1). All other BMTL hyperparameters are held fixed.

TABLE 11. Base learner comparison: BMTL R^2 on agricultural dataset (Mean ± Std, 10 Seeds). Target-only R^2 = 0.393 for all learners.

Base Learner	BMTL R^2	△ over Target-Only
Ridge (primary)	0.536±0.017	+0.143
Random Forest	0.549±0.015	+0.156
GBR	**0.558**±0.013	**+0.165**
SVR-RBF	0.531±0.022	+0.138

BMTL's improvement is consistent across all base learners (ΔR^2 = +0.138 to +0.165), confirming the framework is not specific to Ridge regression. GBR achieves the highest absolute R^2 (0.558) on the agricultural dataset. We retain Ridge as the primary reported model for reproducibility and interpretability, but recommend GBR for practitioners prioritizing predictive performance.

VII. CASE STUDY: APPLE QUALITY PREDICTION

A. PROBLEM SETTING AND DATA

Predicting fruit quality (Brix sugar content) for new apple varieties in Korea is difficult: Citrus farms in Jeju have 2,527 records spanning 2003–2024, but apple data from mountainous regions is limited to 366 records. Table 12 summarizes dataset characteristics.

TABLE 12. Dataset characteristics for agricultural case study.

Crop	Samples	Years	Varieties	Regions
Apple (Target)	366	2015–2024	2	12
Citrus	2,527	2003–2024	5	14
Grape	473	2015–2024	5	8
Pear	355	2015–2024	3	10

We integrated 20,896 daily weather records from the KMA. Adding weather features improved within-domain prediction for citrus by +6.0% (R^2: 0.574 → 0.634).

B. TRANSFER RESULTS

When applying BMTL to predict apple quality using other crops as sources:

1) **Direct transfer fails**: Training on citrus and testing on apple yields $R^2 = -17.77$, indicating the model is worse than predicting the mean. The feature distributions are too different for zero-shot transfer.
2) **Fine-tuning helps partially**: Pre-training on all sources and fine-tuning on apple achieves $R^2 = 0.298$—still below the target-only baseline (0.393), as the strong source initialization biases the model.
3) **BMTL succeeds**: With balance-aware weighting, BMTL achieves $R^2 = 0.536$—a +0.143 R^2 improvement (+36.4% relative) over target-only training.

These results highlight a common pattern in cross-crop transfer: direct application without adaptation fails, but properly balanced transfer can substantially improve predictions.

C. ROBUSTNESS ANALYSIS

1) SENSITIVITY TO RANDOM SEEDS

BMTL shows consistent improvements across all 10 random seeds, with standard deviation of R^2 scores below 0.03—significantly lower than target-only training (std=0.021), suggesting that transfer learning with balanced sampling also reduces variance.

2) DIFFERENT TARGET DOMAINS

We repeat the agricultural experiments with each crop as the target domain. BMTL consistently outperforms baselines regardless of target choice, though the margin varies based on domain characteristics. Citrus as target (with abundant data) shows smaller BMTL gains, consistent with our theory that balance is most critical for small targets.

3) FEATURE SUBSETS

We test BMTL with different feature subsets (quality features only, weather features only, combined). The best performance is achieved with combined features, validating the importance of weather data integration. Weather-only features alone achieve $R^2 = 0.18$ on apple—lower than quality features alone (0.31)—but the combination reaches 0.536, suggesting strong complementarity.

D. FEATURE IMPORTANCE ANALYSIS

The most transferable features across crops are:

- **Weight-to-sugar ratio**: Heavier fruits tend to have lower sugar concentration across all crops—a universal biophysical relationship.
- **Seasonal temperature**: Accumulated heat units (growing degree days) correlate strongly with sugar development across all fruit types.
- **Harvest timing**: Later harvests generally produce sweeter fruit, regardless of variety or species.

These shared biophysical patterns explain why cross-crop transfer is possible despite significant domain differences in absolute sugar content ranges.

VIII. DISCUSSION

A. WHEN DOES BMTL HELP?

1) **Small target data** ($n_T \leq 100$): +117.8% at n=50.
2) **Related but distinct domains**: Sources share structural properties but have distribution shift.
3) **Severe size imbalance** ($n_S \gg n_T$): Balance control prevents source domination.

B. WHEN DOES BMTL HURT?

1) **Abundant target data** ($n_T \geq 500$): −13.6% at n=500. Variance reduction no longer outweighs bias.
2) **Moderate size imbalance**: Instance-level reweighting (Instance-Wt) can outperform BMTL when imbalance is mild (Concrete dataset).
3) **High domain shift**: Even BMTL underperforms target-only on Energy (MMD=0.223). Transfer should be avoided when domain shift is extreme.

C. PRACTICAL GUIDELINES

1) Use BMTL when $n_T < 100$ and $n_S \gg n_T$.
2) Grid search balance ratio over {0.3, 0.5, 0.8}; select by validation performance.
3) Do not rely on domain distance alone—empirically evaluate transferability.
4) Consider source diversity; balance contributions to prevent domination.

D. BROADER SIGNIFICANCE AND APPLICATIONS

1) AGRICULTURAL AND FOOD SECURITY IMPACT

The ability to predict fruit quality for new varieties without years of data collection has direct economic and food security implications:

- **Accelerated variety deployment**: New climate-resilient varieties can be integrated into AI-driven supply chains within 1–2 growing seasons rather than 5–10 years.
- **Regional knowledge transfer**: Models trained in data-rich regions (Jeju citrus, 2003–2024) can be immediately applied to newly cultivated areas with minimal local data, supporting Korea's smart farm expansion into rural regions.
- **Cost reduction**: BMTL's +36.4% relative improvement means fewer destructive physical measurements (laboratory Brix analysis costs approximately $5–10 per sample) are required to achieve equivalent prediction accuracy—directly reducing operational costs for growers.
- **Supply chain optimization**: More accurate early-season quality predictions enable better harvest timing, pricing, and export decisions, reducing food waste estimated at 20–30% for fresh produce in Korea [36].

2) ALIGNMENT WITH NATIONAL AGRICULTURAL AI STRATEGY

Korea's "Digital Agriculture Vision 2030" and the Rural Development Administration's smart farm program target AI-based quality prediction for all major fruit varieties by 2030. BMTL's ability to bootstrap quality prediction for data-scarce new varieties directly supports this roadmap, enabling knowledge reuse from established monitoring programs without costly parallel data collection infrastructure.

3) GENERALIZABILITY BEYOND AGRICULTURE

The source overwhelming problem is not unique to agriculture. It arises in any domain where:

- A target application is new (limited historical data) but related domains are well-studied
- Data collection is expensive, time-consuming, or requires specialized expertise

Examples include: predicting outcomes for a new drug in clinical trials (leverage existing drug databases), energy consumption for new building types (leverage existing building portfolios), and quality control for new manufacturing processes (leverage similar production lines). Our theoretical

framework and the BMTL algorithm apply directly to all such scenarios, making this work broadly relevant to the machine learning for science community.

E. REGIME ANALYSIS: WHEN TO BALANCE VS. WHEN TO SELECT

The cross-crop LOCO and K-source analyses in Sections VI-G–VI-H reveal a regime structure that has important practical implications. The decision between *multi-source balance control* and *single-source selection* depends primarily on the amount of available target data.

1) LARGE-TARGET REGIME ($N_T \geq 200$)

When target data is plentiful, BMTL's multi-source balance control provides its intended benefit. All source crops contribute complementary information, and BMTL's balance-aware weighting prevents any one large source (e.g., Citrus with 2,022 samples) from dominating the gradient. Transfer gains are modest in absolute terms—the target-only baseline is already strong—but BMTL consistently avoids the catastrophic negative transfer exhibited by naive pooling.

2) INTERMEDIATE REGIME ($N_T = 50$–200)

This is BMTL's core operating zone. The target has enough data to provide meaningful validation signal for transferability estimation, but insufficient data to train reliably without transfer support. The +117.8% improvement at n_T=50 and +79.9% at n_T=100 reported in Section VI fall in this regime. Using a conservative balance ratio ($r \leq 0.5$) is recommended to prevent source bias from overwhelming the target signal.

3) FEW-SHOT REGIME ($N_T < 50$)

As demonstrated in Table 8, combining multiple sources in the extreme few-shot regime is counterproductive. Source selection—identifying and using only the single best-transferring source—outperforms multi-source BMTL. The transferability estimation component of BMTL can still be used to identify the best source, but the combination step should be skipped.

Table 13 summarizes the recommended strategy by regime.

TABLE 13. Recommended transfer strategy by target data regime.

Regime	Recommended Strategy	Ratio Setting
$n_T < 50$	Single-source selection	$r = 0.5$ (fixed)
$n_T = 50$–200	BMTL (multi-source)	$r \leq 0.5$, validation-tuned
$n_T > 200$	BMTL (multi-source)	r validation-tuned

This regime structure also explains an apparent inconsistency in our results: BMTL improves strongly on the full-data agricultural benchmark (Table 3, n_T=366, $R^2 = 0.536$) but struggles in the LOCO few-shot setting for Apple and Pear when target-only data is small. The determining factor is not the crop itself but the amount of target data relative to the source pool size.

4) CONNECTION TO DOMAIN ADAPTATION THEORY

From the generalization bound (Theorem 2), the source-weighted divergence term $\sum_k w_k d(S_k, T)$ is fixed by the crop geometry and cannot be reduced by choosing K. However, n_{eff} grows with increasing K (more total source samples are available). In the intermediate and large-target regimes, this n_{eff} growth improves the Rademacher complexity term faster than the divergence cost worsens—so more sources help. In the few-shot regime, n_{eff} grows slowly relative to the divergence cost, and the budget dilution effect dominates. This theoretical picture precisely matches the empirical regime transitions in Table 13.

5) CONNECTION TO CURRICULUM LEARNING

The regime framework also suggests a curriculum strategy: begin training with only the best single source (low noise, low conflict), then progressively add additional sources as target data accumulates through active collection. This curriculum matches the temporal reality of agricultural AI deployment: a new crop variety begins with a single growing season of records and gains more over time. BMTL's transferability estimation can be re-run each season to reassess which additional sources are worth incorporating as the target transitions from the few-shot to intermediate regime.

6) GENERALIZING THE REGIME BOUNDARIES

The thresholds in Table 13 (n_T=50 and n_T=200) are derived empirically from the four-crop Korean fruit dataset. In settings with higher domain shift (larger MMD between all crop pairs), the few-shot threshold may shift upward—requiring more target data before multi-source combination becomes safe. Conversely, in settings with very low domain shift (closely related domains), multi-source BMTL may be beneficial even at n_T=20. A principled approach to setting the regime threshold is to compute the transferability estimates and reject multi-source combination whenever the estimated BMTL R^2 on the validation split is lower than target-only. This self-monitoring criterion is already implementable with the existing BMTL pipeline and requires no additional computation beyond what Algorithm 1 already performs.

7) RELATIONSHIP TO ACTIVE LEARNING

The regime analysis has a natural connection to active learning for transfer. When the system is in the few-shot regime, the highest-value action is to collect more target samples (reducing BMTL's balance constraint). When in the intermediate regime, the highest-value action shifts to source curation: identifying and removing noisy sources that increase the divergence term. This dual strategy—collect target data until BMTL is safe, then curate sources—provides a practical roadmap for deploying transfer learning systems in production agricultural settings where data collection costs must be balanced against prediction accuracy requirements.

F. LIMITATIONS AND FUTURE WORK

- **Regression focus**: Our analysis focuses on regression tasks. Classification scenarios may exhibit different transfer dynamics due to discrete label spaces.
- **Validation data requirement**: The optimal balance ratio requires some target validation data. In extremely data-scarce scenarios ($n_T < 20$), cross-validation may be unreliable.
- **Linear model assumption**: Our theoretical analysis assumes linear models. Deep learning scenarios may require different analysis frameworks and different balance ratio strategies.
- **Computational cost**: Transferability estimation requires training one model per source, scaling linearly with K. For large K (e.g., >100 sources), efficient approximations may be needed.
- **Static ratio**: The current framework uses a fixed balance ratio throughout training. Adaptive or curriculum-based ratio scheduling could further improve performance.

Future work could explore: (1) adaptive ratio selection without validation data, (2) extension to deep neural network architectures, (3) online learning where target data arrives sequentially, (4) multi-task formulations jointly optimizing across all domains, and (5) applications to other agricultural domains such as rice yield prediction and soil quality assessment.

IX. CONCLUSION

We introduced Balanced Multi-Source Transfer Learning (BMTL), a framework for addressing the source overwhelming problem in multi-source transfer learning. Key findings:

A. PROBLEM IDENTIFICATION

Naive multi-source combination is highly inconsistent: improving results in some domains (+23.7% on Agriculture) while causing catastrophic failure in others (R^2 drops from 0.553 to −0.494 on Energy; from −0.392 to −4.187 for Apple in the full LOCO setting). This *source overwhelming* phenomenon is systematic and predictable based on the source-to-target size ratio.

B. METHODOLOGICAL CONTRIBUTION

The three-step BMTL framework—transferability estimation, balance-aware weighting, adaptive sampling—is computationally efficient (6× faster than CORAL, Algorithm 1) and requires only a small validation split of target data. The balance ratio r is the key hyperparameter, and a simple grid search over $\{0.3, 0.5, 0.8\}$ is sufficient in practice.

C. KEY INSIGHT

Source-target balance matters more than domain similarity. Grape (MMD=0.194, highest distance) transfers better than Citrus (MMD=0.074, closest) when Apple is the target, because Grape's moderate size (473 samples) naturally maintains better balance with the small target (366 samples). This finding challenges the conventional wisdom in domain adaptation that "closer sources transfer better" and should inform source selection strategies in practical deployments.

D. REGIME STRUCTURE.

The cross-crop LOCO and K-source analyses reveal that the optimal strategy depends on target data availability. In the few-shot regime ($n_T < 50$), source *selection* outperforms multi-source combination. In the intermediate regime ($n_T = 50$–200), BMTL provides its strongest benefits. In the large-target regime ($n_T > 200$), BMTL avoids catastrophic failures while providing modest gains. Practitioners should diagnose the regime before applying BMTL.

E. EMPIRICAL RESULTS

BMTL improves R^2 over the target-only baseline in 7 of 8 tested scenarios, with gains of +0.143 points (+36.4% relative) on full agricultural data and up to +117.8% on small targets (n=50). On high-shift benchmarks, BMTL is the only transfer-based method that consistently avoids catastrophic negative transfer. On the cross-crop LOCO benchmark, BMTL with tuned ratio is the only method other than target-only that avoids catastrophic R^2 degradation for both Apple and Pear.

F. WHEN BMTL DOES NOT HELP

BMTL provides limited benefit in three identifiable conditions: (i) the target domain is already well-represented by a single dominant source (e.g., Citrus as target, where the large target-only baseline is already strong); (ii) all source domains have high MMD relative to the target (e.g., Pear as target, where all sources are weakly compatible); and (iii) target data is sufficient for reliable standalone training ($n_T > 200$). Practitioners should validate that BMTL outperforms target-only on the held-out validation set before deployment.

G. AGRICULTURAL SIGNIFICANCE

The ability to predict fruit quality for new crop varieties without years of dedicated data collection has direct implications for Korea's smart farm expansion program. BMTL enables AI-based quality prediction systems to become operational within 1–2 growing seasons rather than 5–10 years, reducing the economic and logistical barriers to adopting precision agriculture for newly registered varieties.

The source overwhelming problem we identify likely affects a wide class of practical ML applications where new targets have scarce data but related rich-data domains exist. The regime analysis framework in Table 13 provides actionable guidelines applicable beyond agriculture: to new drug candidates in clinical trials, energy consumption in new building types, and quality control in new manufacturing processes. We hope this work both provides a practical tool for practitioners in data-scarce domains and inspires further theoretical research on the interplay between balance, selection, and combination in multi-source transfer learning.

APPENDIX A
PROOF DETAILS

A. PROOF OF THEOREM 1 (BIAS-VARIANCE TRADEOFF FOR MIXED TRAINING)

Proof: The expected squared error decomposes as:

$$\mathbb{E}[(y - \hat{f}_r(x))^2] = \text{Bias}^2(\hat{f}_r) + \text{Var}(\hat{f}_r) + \sigma^2 \quad (21)$$

Bias Term: Training with ratio r approximately minimizes $r \cdot \mathbb{E}_T[\ell] + (1-r) \cdot \mathbb{E}_S[\ell]$, yielding $f_r^* \approx r \cdot f_T^* + (1-r) \cdot f_S^*$. Thus:

$$\|f_r^* - f_T^*\| = (1-r) \cdot \|f_S^* - f_T^*\| \leq (1-r)\sqrt{C_1} \cdot d(S,T) \quad (22)$$

Squaring: $\text{Bias}^2(r) \leq (1-r)^2 C_1 d(S,T)^2$.

Variance Term: The effective sample size for target-relevant learning is n_T/r, so $\text{Var}(\hat{f}_r) \leq C_2 r/n_T$ by Lemma 2. □

B. PROOF OF THEOREM 2 (BMTL GENERALIZATION BOUND)

Proof: For mixture distribution $\mathcal{D}_{mix} = \sum_k w_k \mathcal{D}_{S_k}$, the triangle inequality gives $d(\mathcal{D}_{mix}, T) \leq \sum_k w_k d(S_k, T)$. With BMTL's balanced sampling ($\beta_k = r^*$), $n_{eff} = n_T + r^* \sum_k n_{S_k}$. The bound follows from standard Rademacher complexity analysis. □

APPENDIX B
ADDITIONAL EXPERIMENTAL DETAILS

A. DATASET STATISTICS

Table 14 provides detailed statistics for all datasets.

TABLE 14. Detailed dataset statistics.

Dataset	Samples	Features	Range	Std
Agricultural Quality (Brix sugar content)				
Apple	366	12	[8.2, 16.8]	1.84
Citrus	2,527	12	[7.1, 14.9]	1.62
Grape	473	12	[12.4, 22.1]	2.31
Pear	355	12	[9.8, 15.2]	1.28
Weather Data (daily records)				
Apple regions	4,392	6	–	–
Citrus regions	11,374	6	–	–
Grape regions	1,903	6	–	–
Pear regions	3,227	6	–	–
UCI Wine Quality				
White wine	4,898	11	[3, 9]	0.87
Red wine	1,599	11	[3, 8]	0.81

B. FEATURE DESCRIPTIONS

Agricultural Quality Features (12 total): Weight (g), sugar content (Brix), acidity (%), firmness (kg/cm^2), color index, diameter (mm), height (mm), harvest date (Julian day), variety code (one-hot), region code, year, growing degree days.

Weather Features (6 total): Daily average temperature (°C), max temperature (°C), min temperature (°C), precipitation (mm), sunshine hours, relative humidity (%).

C. WEATHER DATA INTEGRATION PROCEDURE

Daily weather records were obtained from the Korea Meteorological Administration (KMA) Open Data Portal for the 2003–2024 period. Integration proceeded in three steps:

1) STEP 1 — STATION MATCHING

Each orchard location (identified by region code in the quality dataset) was matched to the nearest KMA weather station using geodesic distance. For apple orchards in Gyeonggi and Gyeongbuk provinces, the median station distance was 12.3 km (max 31 km). For citrus orchards in Jeju, the nearest station was always ≤8 km.

2) STEP 2 — TEMPORAL ALIGNMENT

Quality measurements are recorded at harvest (one record per growing season per orchard). We aggregated daily weather records into a set of seasonal summaries: (i) the 30-day pre-harvest window average (captures ripening conditions), (ii) the full growing season (Julian day 90–harvest date) average, and (iii) accumulated growing degree days (base 10°C). The 30-day window yielded the strongest within-domain correlations with Brix (Pearson r=0.38 for citrus) and was used as the primary weather feature.

3) STEP 3 — MISSING VALUE HANDLING

Days with missing KMA records (<0.3% of records) were imputed using linear interpolation from adjacent days at the same station. Orchard-years with more than 10 consecutive missing days were excluded (affected <15 samples across all crops).

The resulting dataset contains 20,896 daily records linked to 3,721 quality observations. All weather features are standardized (zero mean, unit variance) before model training.

D. HYPERPARAMETER SETTINGS

See Table 15.

TABLE 15. Hyperparameter settings for all methods.

Method	Parameter	Value
Ridge (all)	α (regularization) Solver	1.0 SVD
TCA	Subspace dimension Kernel	10 RBF (γ=auto)
CORAL	λ (regularization) Alignment	1.0 Full covariance
BMTL	Balance ratio r^* ϵ (transferability) Validation split	0.5 (default) 0.01 0.2

E. EXTENDED FEW-SHOT ANALYSIS

Table 16 reports per-crop BMTL performance at $n_T \in \{20, 50\}$ alongside the target-only baseline. These results complement the aggregate statistics in Table 8 and reveal

TABLE 16. Extended few-shot results per target crop (Mean R^2, 10 Seeds). BMTL uses all three remaining crops as sources with fixed r=0.5.

Target	Method	n_T=20	n_T=50
Apple	Target-only	−0.378	−0.284
	BMTL	−0.611	−0.715
Citrus	Target-only	0.286	0.385
	BMTL	−0.263	0.059
Grape	Target-only	0.437	0.515
	BMTL	**0.539**	**0.563**

which crops benefit from transfer in the extreme few-shot setting.

1) GRAPE: CONSISTENT FEW-SHOT BENEFICIARY

Grape is the only crop for which BMTL consistently outperforms target-only at both n_T=20 (+0.102) and n_T=50 (+0.048). This aligns with the full-data LOCO results (Table 7): Grape occupies a favorable position in the feature space—similar enough to the other crops for knowledge transfer, yet small enough that source domination is limited even under naive pooling.

2) APPLE AND CITRUS: MULTI-SOURCE BMTL HURTS IN THE FEW-SHOT REGIME

For Apple, BMTL with all three sources produces notably worse R^2 than target-only at both sample sizes (−0.611 vs. −0.378 at n_T=20; −0.715 vs. −0.284 at n_T=50). The direction of harm *increases* with n_T—unusual behavior explained by Citrus's growing contribution as more target samples provide a better validation signal that still favors Citrus, whose large size then overwhelms the small target. For Citrus as target, multi-source BMTL also underperforms target-only at n_T=20 (−0.263 vs. 0.286), partially recovering at n_T=50 (0.059 vs. 0.385) but never reaching target-only performance.

3) TRANSITION POINT

The BMTL advantage for Grape appears even at n_T=20, suggesting that the transition from "select" to "balance" regime (Table 13) depends on the specific crop pair and source distribution, not just a universal threshold. Practitioners should use the transferability estimation step of BMTL—with leave-one-out source evaluation on the available n_T samples—as a diagnostic to determine whether multi-source combination is beneficial before committing to full BMTL.

4) IMPLICATIONS FOR DATA COLLECTION STRATEGY

The few-shot results suggest that the marginal value of additional target data is high in the n_T=20–50 range for crops where multi-source BMTL underperforms target-only (Apple, Citrus). For Apple: increasing from n_T=20 to n_T=50 barely moves the target-only baseline (−0.378 → −0.284) but does not help BMTL either—implying that the fundamental feature gap (no weather features) is the binding constraint, not sample size. For Citrus: increasing n_T from 20 to 50 improves both target-only (0.286 → 0.385) and BMTL (−0.263 → 0.059), with BMTL's gap closing. These crop-specific trajectories suggest that for Citrus-like crops (large eventual dataset, moderate domain shift), a data collection strategy that prioritizes reaching n_T=100 before deploying BMTL would be more efficient than deploying BMTL immediately with n_T=20.

Table 16 thus serves as a practical reference for agricultural AI practitioners deciding when to activate cross-crop transfer for a newly registered variety: check whether BMTL exceeds target-only on held-out validation data, and if not, continue with target-only training until additional growing seasons provide sufficient data.

F. EXTENDED RESULTS

See Table 17.

TABLE 17. Extended results with multiple metrics (agricultural and wine).

Method	R^2	MAE	RMSE
Agricultural (Apple target)			
Target-only	0.393	1.124	1.456
Naive	0.486	0.982	1.289
Fine-tuning	0.298	1.198	1.534
CORAL	0.387	1.134	1.467
TCA	−0.377	1.892	2.341
BMTL	**0.536**	**0.912**	**1.198**
Wine Quality (Red target, n=100)			
Target-only	0.188	0.521	0.654
Naive	0.110	0.578	0.723
CORAL	0.206	0.498	0.632
BMTL	**0.287**	**0.456**	**0.578**

G. DOMAIN DISTANCE MEASUREMENTS

See Table 18.

TABLE 18. MMD (maximum mean discrepancy) Between all domain pairs.

	Apple	Citrus	Grape	Pear
Apple	0.000	0.074	0.194	0.142
Citrus	0.074	0.000	0.231	0.118
Grape	0.194	0.231	0.000	0.187
Pear	0.142	0.118	0.187	0.000

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